

Proofs and Problem Solving

Week 4: More Limits

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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7. The Monotone Convergence Theorem

Last week we studied how to formally show that a sequence (x_n) converges to a specific limit L , and we developed various rules for manipulating limits. What if we are given a sequence (x_n) and want to show it converges but we *don't* know what the limit is?

This week we will talk about the monotone convergence theorem which says that any increasing sequence which is bounded above must converge to something. We'll then give some examples illustrating how this result can be useful and shed light on something we are already familiar with: decimal expansions. Lastly we will introduce the notion of infinite series.

7.1 The Monotone Convergence Theorem

Definition 7.1 A sequence $(a_n)_{n=1}^{\infty}$ is *increasing* if $a_{n+1} \geq a_n$ for all n . It is *decreasing* if $a_{n+1} \leq a_n$ for all n . We say (a_n) is *monotonic* if it is either increasing or decreasing.

Exercise: Give an example of an increasing sequence. Give an example of a sequence which is not monotonic. Give an example of a sequence which is both increasing and decreasing.

Monotonic sequences are especially important for us because we have:

Theorem 7.1 — Monotone Convergence Theorem (MCT). Let (a_n) be an increasing sequence of real numbers that is bounded above (i.e. there is M so that $a_n \leq M$ for all n). Then (a_n) converges to some limit. If (a_n) is a decreasing sequence and is bounded below, then (a_n) converges to some limit.

Proof. We begin with the first part. Assume that (a_n) is increasing and bounded above. Notice that we are not told what its limit is going to be. This has to come out of our argument somehow, and the first job is to find a candidate for it.

Since (a_n) is bounded above, this means that the set¹ $\{a_n \mid n \in \mathbb{N}\}$ is bounded above.

¹Note the distinction between the sequence (a_n) , in which the order of members is important, and the set $\{a_n \mid n \in \mathbb{N}\}$, in which the order of members is not important or even recorded.

Thus, by the Completeness Axiom, the set has an LUB, call it L . We claim that

$$\lim_{n \rightarrow \infty} a_n = L.$$

We begin the proof of the claim. Let $\epsilon > 0$, we wish to show there is $N \in \mathbb{N}$ so that

$$|a_n - L| < \epsilon$$

for all $n > N$, or equivalently,

$$L - \epsilon < a_n < L + \epsilon. \quad (7.1.1)$$

Since L is an upper bound for $\{a_n \mid n \in \mathbb{N}\}$, we know $a_n \leq L < L + \epsilon$ for all n , thus dealing with the second inequality in (7.1.1). Therefore, it suffices to show that there is $N \in \mathbb{N}$ so that for all $n > N$,

$$L - \epsilon < a_n.$$

Since L is the LUB for $\{a_n \mid n \in \mathbb{N}\}$, this means that $L - \epsilon$ is not an upper bound for this set, so there exists an integer N so that $a_N > L - \epsilon$. Since the sequence is increasing, this means that for all $n > N$,

$$a_n \geq a_N > L - \epsilon.$$

This shows (7.1.1) holds for all $n > N$ and completes the proof.

We leave the proof of the second part to the reader. (Why is it an easy consequence of the first part?) ■

Question: What if a sequence (a_n) is bounded above but only satisfies $a_{n+1} \geq a_n$ for all $n > 100$. Can you still conclude that it converges?

Thus, if you can show that a sequence is bounded and increasing, you can deduce that it has a limit, and in some cases this, together with the rules for limits that we discussed last week, allows you to figure out what the limit actually is.

Example 7.1 Let $a_n = \frac{n-1}{n}$. Then (a_n) is increasing since $\frac{n}{n+1} \geq \frac{n-1}{n} \iff n^2 \geq (n+1)(n-1) = n^2 - 1$, which is true for all n . (a_n) is also bounded above by 1. The MCT tells us that (a_n) converges to some $L \leq 1$.

(In fact we don't need the MCT for this example because $a_n = 1 - \frac{1}{n} \rightarrow 1 - 0 = 1$ by the rules for limits, but it nevertheless illustrates the general point.)

The next example can also be done directly, but demonstrates a very useful general technique: *finding a simple equation which any potential limit must satisfy*.

Example 7.2 If $a_n = t^n$ for some fixed $t \in (0, 1)$, show that the sequence $(a_n)_{n=1}^\infty$ converges and $\lim_{n \rightarrow \infty} a_n = 0$.

Solution: Since t^n is a product of positive numbers, it is positive, so the sequence $(a_n)_{n=1}^\infty$ is bounded below by 0. Moreover, since $t < 1$, $t^{n+1} = t \cdot t^n < t^n$, so the sequence is decreasing. By the MCT, it has a limit L .

To find L , we find a simple equation which L must satisfy. Recall that if (a_n) is a convergent sequence, then (a_{n+1}) also converges, and indeed converges to the same limit (see the exercises from last week). Thus, by the rules for limits,

$$L = \lim_{n \rightarrow \infty} t^n = \lim_{n \rightarrow \infty} t^{n+1} = \lim_{n \rightarrow \infty} t \cdot t^n = t \cdot \lim_{n \rightarrow \infty} t^n = tL.$$

Therefore $L(t-1) = 0$ and since $t \neq 1$ we must have $L = 0$, and we are done.

In this example, we used the rules for limits, together with the fact that if (x_n) converges, then (x_{n+1}) also converges to the same limit. In the next example we instead use the fact that if $x_n \rightarrow L$, then $x_{2n} \rightarrow L$ too (see Exercise 7.4).

Example 7.3 Let $a_n = 2^{\frac{1}{n}}$. Show that (a_n) converges and find its limit.

Solution: We notice that $2^{\frac{1}{n}} \geq 1$ for all n (since this is equivalent to $2 \geq 1^n = 1$) and also that $2^{\frac{1}{n+1}} \leq 2^{\frac{1}{n}}$ for all n (since this is equivalent to $2^n \leq 2^{n+1}$). Therefore (a_n) is a decreasing sequence which is bounded below by 1. Consequently, by the MCT, it converges to some number L which satisfies $L \geq 1$ (and $L < 2$).

Consider the sequence $b_n = a_{2n} = 2^{\frac{1}{2n}}$. It runs through alternate values of the sequence (a_n) , and so $b_n \rightarrow L$ as $n \rightarrow \infty$. But $b_n^2 = a_n$, so by the rules for limits we must have $L^2 = L$, or $L(L-1) = 0$, meaning either $L = 0$ or $L = 1$. Since $1 \leq L < 2$ the only possibility is $L = 1$.

Sometimes it can be helpful to find the possible limits L *before* establishing monotonicity and boundedness. Indeed, knowing possible values of L can help us to know what we are looking for in seeking to establish monotonicity and boundedness.

7.2 Iteratively defined sequences

Iteratively or recursively defined sequences specify the value of x_{n+1} in terms of previous values of the sequence $x_1, \dots, x_n$, (usually just in terms of x_n).

Example 7.4 You may have already seen recursively defined sequences of the form,

$$\begin{cases} x_1 &= c \\ x_{n+1} &= ax_n + b \text{ for } n \geq 1 \end{cases}$$

where $a, b, c \in \mathbb{R}$. Show that when $|a| < 1$, (x_n) converges and find its limit. Show that when $|a| > 1$, (x_n) diverges. What happens when $a = \pm 1$?

Solution: Let's first consider what L might be in terms of a, b and c . By the rules for limits, if the sequence (x_n) converges to L , then L *must* satisfy $L = aL + b$, or equivalently $L = \frac{b}{1-a}$. Since we cannot divide by 0, this already shows us that the case $a = 1$ is special.

Indeed, if $a = 1$, then $L = aL + b$ becomes $L = L + b$, and unless $b = 0$ this cannot hold. So when $a = 1$, the sequence diverges in the case that $b \neq 0$, and is the constant sequence $x_n = c$ in the case that $b = 0$.

Now suppose $a \neq 1$. Since we are guessing the limit to be $\frac{b}{1-a}$, we can use the sequence $y_n = x_n - \frac{b}{1-a}$ to re-write the equation $x_{n+1} = ax_n + b$ in the equivalent form

$$\begin{cases} y_1 &= c - \frac{b}{1-a} \\ y_{n+1} &= ay_n \end{cases} \text{ for } n \geq 1.$$

Using induction, we can establish that $y_n = (c - \frac{b}{1-a})a^{n-1}$.

If $|a| > 1$, $|y_n|$ diverges to $+\infty$ and so (x_n) also diverges; if $|a| < 1$, then $y_n \rightarrow 0$ as we have already seen in a previous example, and this is the same as $x_n \rightarrow \frac{b}{1-a}$. The remaining case $a = -1$ we leave as an exercise.

The MCT is especially useful for proving convergence of some (but not all) iteratively or recursively defined sequences.

Example 7.5 In 60AD, the Greek mathematician Hero of Alexandria wanted to approximate $\sqrt{2}$ with rational numbers. He came up with the following recurrence relation, now called *Heron's Method*, for computing $\sqrt{2}$. Let $x_1 = 2$ and for each $n \in \mathbb{N}$, let

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right).$$

Let us imagine that we knew that (x_n) converged to something, L . What must L satisfy? Well, by the rules for limits, since $x_{n+1} \rightarrow L$ too, and assuming that $L \neq 0$, it must satisfy

$$L = \frac{1}{2} \left(L + \frac{2}{L} \right),$$

or $2L = L + \frac{2}{L}$, or $L = \frac{2}{L}$, which is the same as $L^2 = 2$, or $L = \pm\sqrt{2}$. We need to do two things: (i) we need to show that (x_n) converges and (ii) that the limit really is $L = +\sqrt{2}$.

Given that $x_1 = 2 > \sqrt{2}$, it makes sense to try to show that (x_n) is decreasing, and is bounded below by $\sqrt{2}$, which is equivalent to $x_n > 0$ and $x_n^2 \geq 2$ for all n . Once we have this, the MCT will imply that (x_n) does converge to some number L , which must then satisfy $L \geq 0$ and $L^2 = 2$, so $L = +\sqrt{2}$.

First we show that the sequence (x_n) is bounded below by $\sqrt{2}$ which means $x_n^2 \geq 2$ for all $n \in \mathbb{N}$. If $n = 1$ then $x_1 = 2$ and so $x_1^2 = 4 \geq 2$. To show that $x_n^2 \geq 2$ for all $n \geq 2$, we observe that, since $x_n > 0$,

$$x_{n+1}^2 = \left(\frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \right)^2 \geq 2 \iff \left(x_n + \frac{2}{x_n} \right)^2 \geq 8 \iff x_n^2 + \frac{4}{x_n^2} + 4 \geq 8 \iff \frac{(x_n^2 - 2)^2}{x_n^2} \geq 0.$$

Since the last inequality is clearly true, its equivalent inequality $x_{n+1}^2 \geq 2$ must also be true. So, we have shown that $x_n^2 \geq 2$ for all $n \in \mathbb{N}$.

Next, we want to show (x_n) is decreasing, which is equivalent to $x_n - x_{n+1} \geq 0$, so:

$$x_n - x_{n+1} = x_n - \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) = \frac{x_n^2 - 2}{2x_n}.$$

Thus, we just need to show that this last term is non-negative. The x_n are always positive, this is established by induction: $x_1 > 0$, and if $x_n > 0$, then we see that $x_{n+1} > 0$ straight from the definition. Hence, we just need $x_n^2 \geq 2$ for all $n \geq 1$ but we showed this in the last step. Hence, we conclude that the sequence (x_n) is decreasing.

Thus we conclude from the earlier remarks that (x_n) converges to $L = +\sqrt{2}$!

Point to ponder: How might you modify Heron's formula to construct a sequence of rational numbers converging to $\sqrt{7}$ or to $\sqrt{51}$?

Note from the above argument that induction is a really helpful tool to have at our disposal when we are dealing with recursively defined sequences. We see this again in the next example.

The previous example had the feature that the relevant boundedness of the sequence

was immediate: the terms were all positive (hence bounded below by zero). Let's do an example where the relevant boundedness is not so immediate:

Example 7.6 Let $x_1 = 1$ and $x_{n+1} = 1 + \sqrt{x_n}$ for $n \geq 1$. Prove that (x_n) converges and find its limit.

As before we first give ourselves a sneak preview of what the limit L must satisfy if we assume it exists. By the rules for limits we must have

$$L = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} (1 + \sqrt{x_n}) = 1 + \sqrt{L}. \quad (7.2.1)$$

So L must satisfy $(L - 1)^2 = L$, i.e. $L^2 - 3L + 1 = 0$, i.e. $(L - \frac{3}{2})^2 = \frac{5}{4}$, and this equation has solutions $L = \frac{3 \pm \sqrt{5}}{2}$. Both of these are positive but $\frac{3 - \sqrt{5}}{2} < 1 < \frac{3 + \sqrt{5}}{2}$. Now $x_1 = 1$ and it's easily proved by induction that $x_n \geq 1$ for all n . So if (x_n) converges, it *must* converge to $\frac{3 + \sqrt{5}}{2}$.

If we can prove that (x_n) is increasing and is also bounded, the MCT will imply it converges, and by what we have said above, the limit must be $\frac{3 + \sqrt{5}}{2}$.

We prove it is increasing using induction. We want that $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$. For the base case, we have $x_2 = 1 + \sqrt{1} > 1 = x_1$. For the inductive step, assume for a certain $n \in \mathbb{N}$ that $x_{n+1} \geq x_n$. Then

$$x_{n+2} - x_{n+1} = 1 + \sqrt{x_{n+1}} - (1 + \sqrt{x_n}) = \sqrt{x_{n+1}} - \sqrt{x_n}$$

and since $x_{n+1} \geq x_n$ by the inductive hypothesis, we then also have $\sqrt{x_{n+1}} \geq \sqrt{x_n}$ by Proposition 3.5. So the above displayed expression is non-negative, and thus $x_{n+2} \geq x_{n+1}$. This proves the inductive step and hence that (x_n) is increasing.

Now we need to show it is bounded above. Since we expect it to converge to $L = \frac{3 + \sqrt{5}}{2}$, it makes sense to test this L as an upper bound. Let us prove that $x_n \leq L$ for all $n \in \mathbb{N}$ by induction. We can verify the base case $x_1 = 1 \leq L$ is already true. For the inductive step, it turns out that the actual value of L won't be as useful as the identity $L = 1 + \sqrt{L}$ we derived in equation (7.2.1). Assume we have proved $x_n \leq L$ for some n . Then

$$x_{n+1} = 1 + \sqrt{x_n} \leq 1 + \sqrt{L} = L.$$

This proves the inductive step, and hence proves that $x_n \leq L$ for all n . Hence, (x_n) is bounded above and is increasing, so the MCT implies it converges. As we showed above, it must converge to $L = \frac{3 + \sqrt{5}}{2}$.

7.3 Exercises

■ **Exercise 7.1** Suppose that (x_n) is increasing. Show that either (x_n) converges or else (x_n) tends to infinity.

■ **Exercise 7.2** If (x_n) is a bounded sequence and $(x_n + \frac{1}{n})$ is monotonic, does (x_n) converge?

■ **Exercise 7.3** Let (a_n) be a sequence of numbers such that $0 \leq a_n \leq 1$ and let s_n denote the product $a_1 \cdot a_2 \cdots a_n$. Does (s_n) converge?

■ **Exercise 7.4** Using the definition of convergence, show that if $a_n \rightarrow L$ as $n \rightarrow \infty$ and if $b_n = a_{2n}$, then $b_n \rightarrow L$ as $n \rightarrow \infty$.

■ **Exercise 7.5** Let $a_n = n^{1/n}$. Show that (a_n) converges and find its limit. (**Hint:** First prove that $(1 + \frac{1}{n})^n \leq 4$ for all n , and then try to use the MCT. Alternatively, use the formula for $a^n - b^n$ in Exercise 8.1 for well-chosen a and b .)

■ **Exercise 7.6** Let $x_1 = 1$ and $x_{n+1} = \sqrt{1 + x_n}$ for $n \geq 1$. Show that (x_n) converges and compute its limit.

■ **Exercise 7.7** Let $x_1 = \frac{1}{4}$ and $x_{n+1} = \sqrt{x_n(1 - x_n)}$ for $n \geq 1$. Show that (x_n) converges and compute its limit. (Hint: First show that $0 < x_n \leq \frac{1}{2}$ for all n . The AM–GM inequality may come in useful.)

■ **Exercise 7.8** Let $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + \sqrt{x_n}}$ for $n \geq 1$. Show that (x_n) converges and find the quartic equation of which the limit is a root. (When showing it is bounded above, you can try making an educated guess as to what an upper bound might be, try some values out!)

8. Decimals and Series

We are all familiar with decimal representation of numbers. For example, given a rational number $x = \frac{p}{q}$ with $p, q \in \mathbb{N}$ and $0 < p < q$, we can go through the process of dividing q into p to represent it as

$$x = 0.a_1a_2\dots a_n\dots$$

where each a_n is an integer in the set $\{0, 1, \dots, 9\}$. So we are used to writing

$$\frac{1}{2} = 0.5 = 0.50000\dots \text{ and } \frac{2}{3} = 0.66666\dots$$

and we are also familiar with such expressions as

$$\sqrt{2} = 1.414\dots \text{ and } \pi = 3.14159\dots$$

There are also more "random" looking examples such as $0.7264082793684301\dots$ which have no discernible pattern nor immediate purpose in life. But what exactly do we mean by these expansions? In this chapter we explore this question, and give a precise meaning to decimal expansions. We will show that *every* real number has a decimal expansion, and conversely, every decimal expansion gives rise to a real number. Thus, in some sense, decimal expansions provide a complete description of the real numbers. In order to do this, we use the study of sequences which we have already begun, and, in particular, we will make crucial use of the Completeness Axiom for $\mathbb{R}$.

Then we will introduce infinite series and relate them to decimal expansions. You are probably familiar with certain infinite series already, such as the infinite geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1,$$

but we will examine series such as these from a more general perspective. Now is a good time to recall the formula for a finite geometric series:

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

whenever $r \neq 1$. You will learn more about infinite series if you are taking the course Calculus and its Applications. Finally, we introduce the number e and study some of its basic properties.

8.1 Decimals

Given a sequence (a_n) with each $a_n \in \{0, 1, \dots, 9\}$, what exactly do we mean by the decimal

$$0.a_1a_2\dots a_n\dots?$$

We all know the convention that the number a_1 gives the number of tenths, a_2 gives the number of hundredths, a_3 gives the number of thousandths etc. But what do we mean by the " $\dots$ " at the end of the expansion? A good way to make sense of this is to consider the *finite* decimal expansions

$$x_n := 0.a_1a_2\dots a_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n}$$

and declare the infinite decimal expansion

$$0.a_1a_2\dots a_n\dots$$

to be the limit of the sequence (x_n) as $n \rightarrow \infty$. But this naturally poses the question of whether the limit must even exist. Fortunately this is easy: the sequence (x_n) consists of positive numbers, and they satisfy $x_{n+1} \geq x_n$ so that (x_n) is increasing. Finally, using the formula for a finite geometric progression,

$$x_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n} \leq \frac{9}{10} + \frac{9}{100} + \dots + \frac{9}{10^n} = \frac{9}{10} \frac{1 - (\frac{1}{10})^{n+1}}{1 - \frac{1}{10}} \leq \frac{9}{10} \frac{1}{1 - \frac{1}{10}} = 1,$$

so that (x_n) is bounded above by 1. By the MCT, the sequence (x_n) converges to some $x \in \mathbb{R}$ satisfying $0 \leq x \leq 1$. We have thus proved:

Theorem 8.1 Given a sequence (a_n) with $a_n \in \{0, 1, \dots, 9\}$ for all n , the decimal expansion $x = 0.a_1a_2\dots a_n\dots$ given by the limit of the sequence

$$x_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n}$$

defines a real number x satisfying $0 \leq x \leq 1$.

What about the converse – does every real number x with $0 \leq x < 1$ have a decimal expansion? Our intuition and experience tells us that this is so, but now we prove it more formally.

Theorem 8.2 Let $x \in \mathbb{R}$ satisfy $0 \leq x < 1$. Then there is a sequence of integers (a_n) , with $a_n \in \{0, 1, \dots, 9\}$ for all n , so that if we let

$$x_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n},$$

then $x_n \rightarrow x$ as $n \rightarrow \infty$.

Proof. We find the integers a_n inductively. To find a_1 , we split the interval $[0, 1)$ into 10 equal subintervals

$$\left[0, \frac{1}{10}\right), \left[\frac{1}{10}, \frac{2}{10}\right), \dots, \left[\frac{9}{10}, \frac{10}{10}\right).$$

Notice that

$$0 \leq x - \frac{a_1}{10} < \frac{1}{10}.$$

Then x must be in exactly one of these intervals, $\left[\frac{a_1}{10}, \frac{a_1+1}{10}\right)$, for a unique $a_1 \in \{0, 1, \dots, 9\}$.

To find a_2 , we split the interval $\left[\frac{a_1}{10}, \frac{a_1+1}{10}\right)$ into 10 equal subintervals

$$\left[\frac{a_1}{10}, \frac{a_1}{10} + \frac{1}{10^2}\right), \left[\frac{a_1}{10} + \frac{1}{10^2}, \frac{a_1}{10} + \frac{2}{10^2}\right), \dots, \left[\frac{a_1}{10} + \frac{9}{10^2}, \frac{a_1+1}{10}\right).$$

Then x must be in exactly one of these intervals as well, which can be written as $\left[\frac{a_1}{10} + \frac{a_2}{10^2}, \frac{a_1}{10} + \frac{a_2+1}{10^2}\right)$ for some unique $a_2 \in \{0, 1, \dots, 9\}$. Notice that

$$0 \leq x - \left(\frac{a_1}{10} + \frac{a_2}{10^2}\right) < \frac{1}{100}.$$

Continuing in this way, we find for each n a unique integer $a_n \in \{0, 1, \dots, 9\}$ so that

$$x \in \left[\frac{a_1}{10} + \dots + \frac{a_n}{10^n}, \frac{a_1}{10} + \dots + \frac{a_n+1}{10^n}\right)$$

and so that

$$0 \leq x - \left(\frac{a_1}{10} + \dots + \frac{a_n}{10^n}\right) < \frac{1}{10^n}.$$

Then we have

$$0 \leq x - x_n < \frac{1}{10^n},$$

and since $1/10^n \rightarrow 0$ as $n \rightarrow \infty$, we have that $x_n \rightarrow x$ by the squeeze theorem. ■

What about decimal expansions $x = a_0.a_1a_2\dots$ for real numbers outside the interval $[0, 1)$? We look at this in the exercises. If $a_0 \in \mathbb{Z}$ and $a_n \in \{0, 1, 2, \dots, 9\}$ for $n \geq 1$ we write

$$x = a_0.a_1a_2\dots$$

to mean that the sequence (x_n) defined by

$$x_n = a_0 + \frac{a_1}{10} + \dots + \frac{a_n}{10^n}$$

converges to x .

What about uniqueness of decimal expansions? Can a real number x be represented by two distinct decimal expansions? Unfortunately, yes: for example

$$\frac{1}{2} = 0.500000\dots = 0.499999\dots$$

since, by the formula for finite geometric series, $0.499999\dots 99$ (with n 9's)

$$= \frac{4}{10} + \frac{\frac{9}{100}(1 - (\frac{1}{10})^n)}{1 - \frac{1}{10}} \rightarrow \frac{4}{10} + \frac{1}{10} = \frac{1}{2}$$

as $n \rightarrow \infty$. (Which of these two representations does the procedure of Theorem 8.2 spit out?) It turns out that situations like this are essentially the only ones when this happens (one expansion trailing 0's and the other trailing 9's), and there are never more than two decimal expansions that give the same number.

Theorem 8.3 Let $0 \leq x < 1$ and suppose that $x = 0.a_1a_2\ldots = 0.b_1b_2\ldots$. Let ℓ be the smallest integer for which $a_\ell \neq b_\ell$ and suppose that $a_\ell < b_\ell$. Then $b_\ell = a_\ell + 1$, and for all $k > \ell$ we have $b_k = 0$ and $a_k = 9$.

Proof. Assuming $\ell \in \mathbb{Z}$ is the smallest integer such that $a_\ell \neq b_\ell$, then

$$0.b_1b_2\ldots b_\ell 000\ldots \leq 0.b_1b_2\ldots b_\ell b_{\ell+1}b_{\ell+2}\ldots = x$$

and similarly,

$$0.a_1a_2\ldots a_\ell 999\ldots \geq 0.a_1a_2\ldots a_\ell a_{\ell+1}a_{\ell+2}\ldots = x.$$

Since $a_j = b_j$ for $1 \leq j \leq \ell - 1$, we therefore have

$$0.a_1a_2\ldots a_{\ell-1}b_\ell 000\ldots \leq x \leq 0.a_1a_2\ldots a_{\ell-1}a_\ell 999\ldots$$

We can also assume $a_\ell < b_\ell$ which implies that $a_\ell + 1 \leq b_\ell$. However, using the formula for a finite geometric series as before, we know

$$0.a_1a_2\ldots a_{\ell-1}a_\ell 999\ldots = 0.a_1a_2\ldots a_{\ell-1}(a_\ell + 1)000\ldots \geq 0.a_1a_2\ldots a_{\ell-1}b_\ell 000\ldots$$

Therefore $b_\ell \leq a_\ell + 1$. As we are assuming $a_\ell + 1 \leq b_\ell$, this forces $b_\ell = a_\ell + 1$.

Now we have

$$x = 0.a_1a_2\ldots a_{\ell-1}a_\ell a_{\ell+1}\ldots = 0.a_1a_2\ldots a_{\ell-1}(a_\ell + 1)b_{\ell+1}\ldots$$

and if for some $k > \ell$ we had either $a_k < 9$ or $b_k > 0$, this would be violated (see Exercise 8.4). Hence we conclude that for all $k > \ell$ we have $a_k = 9$ and $b_k = 0$. ■

When we find a decimal expansion of a rational number, it begins to repeat. Why is this? Because when doing long division, eventually you obtain some remainder a second time, and then you know the division process will repeat what you did from the beginning. This holds for all rational numbers.

Lemma 8.4 If $x \geq 0$ is rational, then it has a periodic decimal expansion, that is,

$$x = a_0.a_1a_2\ldots a_k \overline{b_1b_2\ldots b_\ell}.$$

Here, this notation means that the digits $b_1, \ldots, b_\ell$ repeat, so

$$a_0.a_1a_2\ldots a_k \overline{b_1b_2\ldots b_\ell} = a_0.a_1a_2\ldots a_k b_1b_2\ldots b_\ell b_1b_2\ldots b_\ell \ldots$$

Before proving this, we will introduce a technique which is very useful throughout mathematics.

Definition 8.1 - Pigeonhole Principle

Let S be a set and suppose $k > n$ are integers with $a_1, \ldots, a_k \in S$ and $|S| = n$. Then there exists $i \neq j$ with $a_i = a_j$.



10 pigeons, 9 boxes.

Colloquially, the pigeonhole principle says that if you are sorting k objects into n sets and $k > n$, then there must be at least one set which contains more than one object.

Proof of Lemma 8.4. Write $x = p/q$ where p and q are positive integers. Each step of the long division algorithm $q \overline{)p.0000\dots}$ returns remainders in $\{0, \dots, q-1\}$. By the *pigeonhole principle*, one remainder, call it k , must occur twice. Because $p.0000\dots$ has trailing zeroes, the algorithm starts repeating at that point. ■

We define the *period* of a decimal expansion to be the smallest number of digits in a repeating sequence in the decimal expansion. For example, $0.121212\dots$ has period 2, $0.3434\overline{5}$ has period 3. If there are no repeating sequences we say that the decimal expansion is *aperiodic*.

Corollary 8.5 If p and q are positive integers, the rational number p/q has a decimal expansion with period at most q .

The converse to Lemma 8.4 is also true:

Lemma 8.6 If $x \geq 0$ has a periodic decimal expansion, then x is rational.

Proof. Suppose that $x = a_0.a_1a_2\dots a_k\overline{b_1b_2\dots b_\ell}$. Let $y = 0.\overline{b_1b_2\dots b_\ell}$. If we show that y is rational, then so will x be, because

$$x = a_0 + \frac{a_1}{10} + \dots + \frac{a_k}{10^k} + \frac{y}{10^k}.$$

Let $z = 0.b_1b_2\dots b_\ell 000\dots$, which is certainly rational. Now $y = \lim_{n \rightarrow \infty} y_n$ where

$$y_n = z \left(1 + \frac{1}{10^\ell} + \frac{1}{10^{2\ell}} + \dots + \frac{1}{10^{n\ell}} \right) = \frac{z(1 - 10^{-(n+1)\ell})}{1 - 10^{-\ell}}$$

so that

$$y = \frac{z}{1 - 10^{-\ell}}$$

which is indeed rational. ■

We now summarise the results on rationality and periodicity:

Theorem 8.7 A number $x \geq 0$ is rational $\iff$ it has a periodic decimal expansion.

Noting that all the x which have two distinct decimal expansions are necessarily rational as one of these expansions is finite, we obtain:

Corollary 8.8 A number $x \geq 0$ is irrational $\iff$ it has an aperiodic decimal expansion.

Note that we proved this without ever working with irrational numbers! This gives us a way of constructing many irrational numbers, not just ones that arise as roots like $\sqrt{2}$ or $3^{\frac{1}{2}}$.

Example 8.1 The number $0.10010001\dots$ where the number of zeros between each 1 is *strictly* increasing is irrational, because there is no way the decimal expansion can repeat.

8.2 Infinite Series

We know how to add up a finite set of numbers $a_1 + a_2 + \cdots + a_n$. But what does it mean to add up an infinite sequence of numbers

$$a_1 + a_2 + \cdots + a_j + \cdots?$$

A good way to make sense of this is to form the *new sequence* (s_n) where

$$s_n := a_1 + a_2 + \cdots + a_n$$

and to define the infinite sum

$$a_1 + a_2 + \cdots + a_j + \cdots = \sum_{j=1}^{\infty} a_j := \lim_{n \rightarrow \infty} s_n$$

to be the limit of the sequence (s_n) whenever it exists. The numbers s_n are called the *partial sums* of $\sum_{j=1}^{\infty} a_j$. If the sequence of partial sums (s_n) does *not* converge, we assign *no* meaning to $\sum_{j=1}^{\infty} a_j$. More formally:

Definition 8.2 Given a sequence (a_j) , we say that the infinite series $\sum_{j=1}^{\infty} a_j$ *converges* if the sequence of partial sums

$$s_n = \sum_{j=1}^n a_j$$

converges as $n \rightarrow \infty$. When (s_n) converges we denote its limit by $\sum_{j=1}^{\infty} a_j$.

Alternatively, given the sequence (a_n) , we can equivalently define the sequence (s_n) recursively by $s_1 = a_1$ and $s_n = a_n + s_{n-1}$ for $n \geq 2$.

Question: Give an example of an infinite series $\sum_{n=1}^{\infty} a_n$ which converges, and an example of one which fails to converge.

Decimal expansions can be thought of as special cases of infinite series: given a sequence (b_j) with each $b_j \in \{0, 1, \dots, 9\}$, we can interpret the decimal expansion $0.b_1b_2\dots$ as the infinite series $\sum_{j=1}^{\infty} \frac{b_j}{10^j}$.

A word of warning about a possible source of confusion. Given a sequence (a_n) , we can consider two questions about it: convergence of the **sequence** (a_n) , and convergence of the **series** $\sum_{j=1}^{\infty} a_j$. **It is important to realise that these are two completely different things.** Convergence of the sequence (a_n) is just what it says, while convergence of the series $\sum_{j=1}^{\infty} a_j$ is defined as convergence of the *auxiliary* sequence (s_n) . So it's always important to be clear whether we are talking about convergence of the sequence (a_n) or convergence of the series $\sum_{j=1}^{\infty} a_j$. Some texts may therefore use the word "summable" to describe convergent series.

Sometimes we will also work with series which start from a different position, like $\sum_{j=2}^{\infty} \frac{1}{j(j-1)} = \frac{1}{2(2-1)} + \frac{1}{3(3-1)} + \cdots$, and the definition is the same: we say this series is convergent if the partial sums $\sum_{j=2}^n \frac{1}{j(j-1)}$ converge. Whether we use the "dummy variable" j or k (or anything else) doesn't matter: $\sum_{j=2}^{\infty} \frac{1}{j(j-1)}$ has exactly the same meaning as $\sum_{k=2}^{\infty} \frac{1}{k(k-1)}$ or $\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$.

Example 8.2 The quintessential example of a convergent series is the *geometric series*. Recall that for any real number $r \neq 1$, we can consider the *finite sum*

$$\sum_{j=0}^n ar^j = \frac{a(1-r^{n+1})}{1-r} \quad (8.2.1)$$

If $|r| < 1$, then $|r^{n+1}| = |r|^{n+1} \rightarrow 0$ by Example 7.2, and so $r^{n+1} \rightarrow 0$ as well. Thus,

$$\lim_{n \rightarrow \infty} \sum_{j=0}^n ar^j = \lim_{n \rightarrow \infty} \frac{a(1-r^{n+1})}{1-r} = \frac{a - a \lim_{n \rightarrow \infty} r^{n+1}}{1-r} = \frac{a}{1-r}.$$

Thus, when $|r| < 1$, $\sum_{j=0}^{\infty} ar^j$ is convergent, and $\sum_{j=0}^{\infty} ar^j = \frac{a}{1-r}$. We'll see below that this series converges if and only if $|r| < 1$.

One useful necessary condition for a series to converge is the following:

Proposition 8.9 Suppose the series $\sum_{j=1}^{\infty} a_j$ converges. Then the sequence (a_j) satisfies $a_j \rightarrow 0$.

Proof. Let $s_n = \sum_{j=1}^n a_j$. Then for some $s \in \mathbb{R}$, $s_n \rightarrow s$. Hence $a_n = s_n - s_{n-1} \rightarrow s - s = 0$. ■

As a consequence of this, we see that the geometric series of Example 8.2 diverges when $|r| \geq 1$ since the sequence $(ar^j)_{j=1}^{\infty}$ does not converge to 0 when $|r| \geq 1$.

Just as we had rules for manipulating limits of sequences, we also have rules for manipulating infinite sums.

Proposition 8.10 Let (a_j) and (b_j) be sequences. If $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ are convergent, then so is $\sum_{j=1}^{\infty} (a_j + b_j)$, and

$$\sum_{j=1}^{\infty} (a_j + b_j) = \sum_{j=1}^{\infty} a_j + \sum_{j=1}^{\infty} b_j.$$

If $c \in \mathbb{R}$, then $\sum ca_j$ is convergent and

$$\sum_{j=1}^{\infty} ca_j = c \sum_{j=1}^{\infty} a_j.$$

Proof. We observe that

$$\begin{aligned} \sum_{j=1}^{\infty} (a_j + b_j) &= \lim_{n \rightarrow \infty} \sum_{j=1}^n (a_j + b_j) \\ &= \lim_{n \rightarrow \infty} \left(\sum_{j=1}^n a_j + \sum_{j=1}^n b_j \right) \\ &= \lim_{n \rightarrow \infty} \sum_{j=1}^n a_j + \lim_{n \rightarrow \infty} \sum_{j=1}^n b_j = \sum_{j=1}^{\infty} a_j + \sum_{j=1}^{\infty} b_j. \end{aligned}$$

Similarly,

$$\sum_{j=1}^{\infty} ca_j = \lim_{n \rightarrow \infty} \sum_{j=1}^n ca_j = c \lim_{n \rightarrow \infty} \sum_{j=1}^n a_j = c \sum_{j=1}^{\infty} a_j.$$

■

Point to ponder: What about a rule for *products* of series?

In Calculus and its Applications, you will learn about several tests for convergence of series. We will only discuss one here.

Theorem 8.11 — Comparison Test. If $b_j \geq 0$ for all j and $\sum_{j=1}^{\infty} b_j$ is convergent, and if $|a_j| \leq b_j$ for all j , then $\sum_{j=1}^{\infty} a_j$ is also convergent, and moreover $|\sum_{j=1}^{\infty} a_j| \leq \sum_{j=1}^{\infty} b_j$.

Proof. The idea is to write $a_j = (a_j - b_j) + b_j$ and to use Proposition 8.10. We know that $\sum_j b_j$ converges by hypothesis, and if we knew that $\sum_j (a_j - b_j)$ converged we would be in business.

Note first that, since $a_j \leq b_j$ for all j , we have $t_n \geq 0$ for all n , and moreover the sequence (t_n) given by

$$t_n = \sum_{j=1}^n (b_j - a_j)$$

is increasing. Next, note that $b_j - a_j \leq 2b_j$, and so

$$t_n \leq \sum_{j=1}^n 2b_j \leq \sum_{j=1}^{\infty} 2b_j$$

and since this latter sum converges by Proposition 8.10, this means that (t_n) is bounded above. Thus, by the MCT, (t_n) converges and $\lim_{n \rightarrow \infty} t_n \geq 0$. Therefore, by Proposition 8.10, $\sum_{j=1}^{\infty} (a_j - b_j)$ converges to some nonpositive number. We conclude that $\sum_{j=1}^{\infty} a_j$ converges, and the sum is $\sum_{j=1}^{\infty} b_j + \sum_{j=1}^{\infty} (a_j - b_j)$, whose value is at most $\sum_{j=1}^{\infty} b_j$. Finally, applying the same reasoning with $-a_j$ in place of a_j , we conclude that

$$\left| \sum_{j=1}^{\infty} a_j \right| \leq \sum_{j=1}^{\infty} b_j.$$

■

For example, since $\left| \frac{\sin j}{2^j} \right| \leq \frac{1}{2^j}$ and $\sum_{j=1}^{\infty} \frac{1}{2^j}$ converges by Example 8.2, we can deduce that $\sum_{j=1}^{\infty} \frac{\sin j}{2^j}$ converges too.

8.3 The number e

The number e is a very important real number. It arose in practical calculations in compound interest when one wants to calculate the interest on a "continuous" basis rather than on a discrete (annually, monthly, daily, hourly) basis. In this context, we often see e as the limit of the sequence (a_n) , where

$$a_n = \left(1 + \frac{1}{n} \right)^n.$$

It was also used as the base for natural logarithms. These were invented as early as 1618 by John Napier in the Merchiston area of Edinburgh.

We can also define e as the infinite sum

$$e := \sum_{n=0}^{\infty} \frac{1}{n!}$$

which converges by the comparison test since for $n \geq 1$

$$\frac{1}{n!} = \frac{1}{n \cdot (n-1) \cdots 2 \cdot 1} \leq \frac{1}{2 \cdot 2 \cdots 2 \cdot 1} = \frac{1}{2^{n-1}},$$

and the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ converges. This is the definition of e which we adopt officially in mathematics. In one of the exercises (quite hard!) we establish that

$$\sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n,$$

showing that the two definitions of e which we have proposed are in fact the same.

What is not clear from either definition of e is whether it is rational or irrational: it is not clear whether e has a repeating decimal expansion or not from its definition. In the Appendix below we establish that e is in fact irrational. In the Honours Analysis course you will study the more general exponential function which gives a rigorous framework for raising the number e to an arbitrary (not just rational) real power x , and much else besides.

8.4 Exercises

The relevant exercises in Liebeck are in Chapter 3.

■ **Exercise 8.1** Use the formula for a finite geometric series to show that for $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$ we have

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1}).$$

■ **Exercise 8.2** Write each of the following decimal expansions in the form of a fraction $\frac{p}{q}$ with $p, q \in \mathbb{Z}$.

(a) $0.\overline{123}$.

(b) $0.\overline{2}$.

(c) $0.\overline{2331}$.

■ **Exercise 8.3** Prove that if $a_n, b_n \in \{0, 1, \dots, 9\}$ are so that $a_n + b_n \leq 9$ for all n , then

$$0.a_1a_2\cdots + 0.b_1b_2\cdots = 0.(a_1 + b_1)(a_2 + b_2)\cdots$$

■ **Exercise 8.4** Suppose that $0 \leq a_1 \leq 8$ and that $0.a_1a_2a_3\cdots = 0.(a_1 + 1)b_2b_3\cdots$. Show that $a_j = 9$ and $b_j = 0$ for all $j \geq 2$.

■ **Exercise 8.5** Suppose that $a_n \geq 0$ and the numbers $\sum_{j=1}^n a_j$ are bounded above. Is $\sum_{j=1}^{\infty} a_j$ convergent?

■ **Exercise 8.6** In each case, either find a sequence (a_n) such that $a_n \geq 0$ for all n which has the desired property, or else explain why no such sequence exists.

(i) the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ both converge

(ii) the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ both fail to converge

(iii) the sequence (a_n) converges and the series $\sum_{n=1}^{\infty} a_n$ fails to converge

(iv) the sequence (a_n) fails to converge and the series $\sum_{n=1}^{\infty} a_n$ converges.

■ **Exercise 8.7** If $\sum_n |a_n|$ converges, show that $\sum_n a_n$ converges. **Harder:** Does the converse hold?

■ **Exercise 8.8** If $a_n \rightarrow 0$, does $\sum_n a_n$ necessarily converge?

■ **Exercise 8.9** Suppose that $|a_n - a_{n-1}| \leq 2^{-n}$ for all $n \in \mathbb{N}$. Show that the sequence (a_n) converges. (**Hint:** Consider the series $\sum_{n=1}^{\infty} (a_n - a_{n-1})$.)

■ **Exercise 8.10** A fun fact you'll learn how to prove if you take the 4th year course *Fourier Analysis* is that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Given this fact, compute $\sum_{n=1}^{\infty} \frac{1}{(2n+1)^2}$.

■ **Exercise 8.11** Let $a_n = \left(1 + \frac{1}{n}\right)^n$.

- (i) Show that $a_{n+1} \geq a_n$ if and only if $\left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \geq \frac{n}{n+1}$.
- (ii) Use Bernoulli's inequality to show that $\left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \geq \frac{n}{n+1}$, and deduce that (a_n) is increasing.
- (iii) Use the binomial theorem and finite geometric series to show that $a_n \leq 4$ for all n .
- (iv) Deduce that (a_n) converges to some number L satisfying $2 < L \leq 4$.

■ **Exercise 8.12** (Quite challenging.) Let $a_n = \left(1 + \frac{1}{n}\right)^n$, and let $b_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$. Show that (a_n) and (b_n) both converge to the same limit. (You may assume that (a_n) is convergent.)

■ **Exercise 8.13** Show that for every real number x there is a unique integer $a_0 \in \mathbb{Z}$ such that $a_0 \leq x < a_0 + 1$. (We sometimes call a_0 the *integer part* of x , or the "floor" or "staircase" of x , denoted by $a_0 = \lfloor x \rfloor$.) Deduce that every real number x has a decimal expansion $x = a_0.a_1a_2\dots a_na_n\dots$ where $a_0 \in \mathbb{Z}$ and $a_n \in \{0, 1, 2, \dots, 9\}$.

■ **Exercise 8.14**

Formulate and prove results analogous to Theorems 8.3 and 8.7 for arbitrary $x \in \mathbb{R}$.

8.5 *Appendix: irrationality of e

This section is optional, and is here to give you a taste of *analytic number theory*.

In addition to Corollary 8.8, there is another method for proving irrationality which involves this rather simple-looking lemma.

Lemma 8.12 Let $x \in \mathbb{R}$. Suppose that for every $c > 0$ there is a rational number $\frac{p}{q} \neq x$ so that

$$\left| x - \frac{p}{q} \right| < \frac{c}{q}. \quad (8.5.1)$$

Then x is irrational.

Proof. We prove this using the contrapositive. Suppose x is rational. Then $x = \frac{a}{b}$ for some integers a and b with $b > 0$. Hence if $\frac{p}{q}$ is any other rational number not equal to x ,

$$\left| x - \frac{p}{q} \right| = \left| \frac{a}{b} - \frac{p}{q} \right| = \left| \frac{aq - pb}{bq} \right| = \frac{|aq - pb|}{bq}.$$

Note that $aq - pb$ is an integer. Moreover $aq - pb \neq 0$, since otherwise we would have $x = p/q$. Hence, $|aq - pb| \geq 1$, and if we set

$$c = \frac{1}{b},$$

then we have that for any rational number $\frac{p}{q}$,

$$\left| x - \frac{p}{q} \right| \geq \frac{|aq - pb|}{bq} \geq \frac{1}{bq} = \frac{c}{q}.$$

We have shown that, supposing x is rational, there exists a $c > 0$ such that for all rational numbers $\frac{p}{q}$ we have $\left| x - \frac{p}{q} \right| \geq \frac{c}{q}$ and we are done. ■

As a corollary, we get the following:

Corollary 8.13 Suppose that $p_n, q_n \in \mathbb{Z}$, $q_n \neq 0$, and $\left| x - \frac{p_n}{q_n} \right| q_n \rightarrow 0$. Then x is irrational.

Proof. Let $c > 0$. By assumption, there is an N so that $n > N$ implies

$$\left| x - \frac{p_n}{q_n} \right| q_n < c$$

or equivalently

$$\left| x - \frac{p_n}{q_n} \right| < \frac{c}{q_n}.$$

Thus, for all $c > 0$, we can find $\frac{p_n}{q_n}$ rational so that (8.5.1) holds. Thus x is irrational. ■

Theorem 8.14 The number e is irrational.

Proof. Let $x = e - 1$. We aim to show that for every $c > 0$, we can find a rational number $\frac{p}{q}$ so that (8.5.1) holds. Lemma 8.12 will then imply that $e - 1$, and hence e , is irrational.

So let $c > 0$. Let $N \in \mathbb{N}$. Then

$$x = \sum_{n=1}^{\infty} \frac{1}{n!} = \sum_{n=1}^N \frac{1}{n!} + \sum_{n=N+1}^{\infty} \frac{1}{n!}$$

Note that

$$\begin{aligned} \sum_{n=1}^N \frac{1}{n!} &= 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{N!} \\ &= \frac{N!}{N!} + \frac{N!/2!}{N!} + \frac{N!/3!}{N!} + \cdots + \frac{1}{N!} \\ &= \frac{N! + \frac{N!}{2!} + \frac{N!}{3!} + \cdots + 1}{N!} = \frac{p}{N!} \end{aligned}$$

where p is an integer since $N!/k!$ is an integer for $k = 0, 1, \dots, N$. We will now show, for N large enough, that

$$\left| x - \frac{p}{N!} \right| < \frac{c}{N!}.$$

We know exactly what $x - \frac{p}{N!}$ is:

$$x - \frac{p}{N!} = x - \sum_{n=1}^N \frac{1}{n!} = \sum_{n=N+1}^{\infty} \frac{1}{n!} = \frac{1}{(N+1)!} \sum_{n=N+1}^{\infty} \frac{(N+1)!}{n!}$$

If we can show that there is some constant C such that

$$\sum_{n=N+1}^{\infty} \frac{(N+1)!}{n!} \leq C$$

for all N , we are done, since then

$$\left| x - \frac{p}{N!} \right| = \frac{1}{(N+1)!} \sum_{n=N+1}^{\infty} \frac{(N+1)!}{n!} \leq \frac{C}{(N+1)!} = \frac{C/(N+1)}{N!}$$

and so if $\frac{C}{N+1} < c$ (which happens if we pick $N > \frac{C}{c} - 1$), the above gives $\left| x - \frac{p}{N!} \right| < \frac{c}{N!}$.

For the missing step, we have that for $n \geq N+1$,

$$\frac{(N+1)!}{n!} = \frac{1}{\binom{n}{N+1}} \frac{1}{(n-N-1)!} \leq \frac{1}{(n-N-1)!}$$

since the binomial coefficients are all nonnegative integers, and we have already seen that

$$\sum_{n=N+1}^{\infty} \frac{1}{(n-N-1)!} = \sum_{n=0}^{\infty} \frac{1}{n!}$$

converges.

To recap, for every $c > 0$, we have shown that we can pick N so that $\left| x - \frac{p}{N!} \right| < \frac{c}{N!}$, which implies that e is irrational by Lemma 8.12. ■